Learning Objective 1.3: I can determine the polarization of an antenna and describe the bandwidth characteristics of common antenna types.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Identify the polarization.** For each plane wave propagating in $+\hat{\mathbf{z}}$, identify the polarization (linear, RHCP, LHCP, or elliptical) and — where linear — the tilt angle relative to $\hat{\mathbf{x}}$.

   (a) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz)$

   Only an $\hat{\mathbf{x}}$ component — **linear along $\hat{\mathbf{x}}$** (0°).

   (b) $\vec{E} = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\cos(\omega t - kz)$

   Equal amplitudes, in phase — **linear, tilted** +45°.

   (c) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz - 90°)$

   Equal amplitudes, $\delta = -90°$, $+\hat{\mathbf{z}}$ propagation — **RHCP**.

   (d) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz + 90°)$

   $\delta = +90°$ — **LHCP**.

   (e) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz) + \hat{\mathbf{y}}1\cos(\omega t - kz - 90°)$

   Unequal amplitudes, $\delta = -90°$ — **right-hand elliptical**, AR = 3 ($\approx$ 9.5 dB).

2. **Axial ratio.** A satellite downlink antenna is spec'd circularly polarized with AR $\leqslant 3$ dB across its operating band.

   (a) Convert the 3 dB axial ratio to a linear ratio (major/minor axis).

   $\mathrm{AR} = 10^{3/20} \approx 1.41$ (i.e. $\sqrt{2}$).

   (b) Write the two orthogonal linear components (up to a common scale) with the appropriate 90° phase, assuming the ellipse major axis is along $\hat{\mathbf{x}}$.

   $$E_x(t) = 1 \cdot \cos(\omega t), \qquad E_y(t) = 0.707\cos(\omega t \pm 90^\circ)$$

   Sign of the phase sets the RHCP vs LHCP sense.

   (c) An $\hat{\mathbf{x}}$ receiver picks off E_x; a $\hat{\mathbf{y}}$ receiver picks off E_y. Compute the ratio of received *power* in dB. Should it match part (a)?

   $$\frac{P_x}{P_y} = \left(\frac{1.0}{0.707}\right)^2 = 2.0 \longrightarrow 10\log_{10}(2.0) \approx 3.0 \text{ dB}$$

   Yes — the axial ratio in dB equals the received-power difference between the two aligned linear receivers.

3. **Polarization loss factor.**

   (a) A ground station transmits a *linearly* polarized signal along $\hat{\mathbf{x}}$. A satellite receiver is **RHCP** and pointed correctly. Compute the PLF in dB.

   Linear $\leftrightarrow$ CP: $\text{PLF} = 0.5 = -3$ dB .

   (b) Same ground station, but the satellite receiver is **LHCP**. Compute the PLF.

   Still -3 dB — linear-to-CP does not depend on sense.

   (c) Two linear whip antennas: radio A vertical, radio B tilted 30° from vertical. Compute the PLF.

   $\text{PLF} = \cos^2(30^\circ) = (0.866)^2 \approx 0.75 \longrightarrow 10\log_{10}(0.75) \approx -1.2$ dB .

   (d) In one sentence, explain why aircraft-to-ground links almost always specify circular polarization on at least one end.

   Airframe attitude (roll/pitch/yaw) sweeps a linear aircraft antenna through every tilt relative to the ground station; making one end CP caps the worst-case polarization loss at 3 dB regardless of attitude.

4. **Impedance bandwidth from a VSWR spec.** An antenna is spec'd "VSWR $\leqslant 2$ from 2.30GHz to 2.55GHz."

   (a) What is the impedance bandwidth in MHz?

   $2.55 - 2.30 = 250\text{MHz}$.

   (b) What is the fractional bandwidth (FBW) in percent?

$$f_c = 2.425\text{GHz}, \qquad \text{FBW} = \frac{250}{2425} \approx 10.3\%$$

   (c) Convert VSWR = 2 to $|\Gamma|$ and to return loss in dB.

$$|\Gamma| = \frac{2-1}{2+1} = \frac{1}{3} \approx 0.333, \qquad \text{RL} = -20\log_{10}(0.333) \approx 9.5 \text{ dB}$$

   (d) Narrowband, broadband, or ultra-wideband? Likely family — patch, horn, or log-periodic?

   FBW $\approx$ 10% is **broadband**: a standard-gain horn or a well-designed half-wave dipole. A patch would need matching tricks; a log-periodic is overkill.

5. **Fractional and ratio bandwidth.**

(a) A cellular antenna covers 698–960MHz (low) and 1.71–2.69GHz (high). Compute the FBW and RBW of each band.

$$\text{Low: } f_c = 829\text{MHz}, \quad \text{FBW} = \frac{262}{829} \approx 31.6\%, \quad \text{RBW} = \frac{960}{698} \approx 1.38{:}1$$
$$\text{High: } f_c = 2200\text{MHz}, \quad \text{FBW} = \frac{980}{2200} \approx 44.5\%, \quad \text{RBW} = \frac{2690}{1710} \approx 1.57{:}1$$

(b) A UWB radar antenna covers 3.1–10.6GHz. Compute its RBW. Does it meet the FCC UWB threshold of RBW $\geqslant$ 2:1?

$$\text{RBW} = \frac{10.6}{3.1} \approx 3.42{:}1$$

Yes — well above the 2:1 threshold.

(c) Which antenna family would you pick for each of the three bands, with a one-sentence reason?

Low ($\approx$ 32%): broadband monopole or PIFA. High ($\approx$ 45%): chassis slot or stacked patch. UWB (3.4:1): Vivaldi (tapered slot) or log-periodic — self-scaling over a wide band without matching tricks.

6. **Chu-Harrington intuition.** A design must fit inside a sphere of radius $a = 15$ mm and operate at $f = 915\text{MHz}$ (ISM band).

   (a) Compute λ, k, and the electrical size ka.

   $$\lambda = \frac{c}{f} \approx 0.328\text{m}, \quad k = \frac{2\pi}{\lambda} \approx 19.2 \text{ rad/m}, \quad ka = 19.2(0.015) \approx 0.288$$

   (b) Estimate the minimum Q from $Q_{\min} \approx \dfrac{1}{(ka)^3} + \dfrac{1}{ka}$.

   $$Q_{\min} \approx \frac{1}{0.0239} + 3.47 \approx 45$$

   (c) Estimate the maximum fractional bandwidth $\text{FBW}_{\max} \approx 1/Q_{\min}$ (for VSWR $\leqslant 2$).

   $\approx 1/45 \approx 2.2\%$ ($\approx 20\text{MHz}$ at 915 MHz).

   (d) The designer wants 100 MHz of usable bandwidth at 915 MHz. Possible in a 15 mm sphere?

   No — 100MHz is FBW $\approx 11\%$, roughly $5\times$ the Chu-Harrington ceiling for a 15 mm sphere. It requires a larger antenna, or resistive loading / matching-network absorption that trades gain for bandwidth.

7. **Mismatch that kills the amplifier.** A power amplifier (PA) delivers a forward power of $P_{fwd} = +40$ dBm toward a transmit antenna. A shunt protection diode at the PA output fails catastrophically if it must dissipate more than $P_{max} = +30$ dBm . Model *all* of the reflected power as landing on the diode.

   (a) Convert P_{fwd} and P_{max} to watts.

   $$P_{fwd} = 10^{(40-30)/10}\text{W} = 10\text{W}\ , \qquad P_{max} = 10^{0}\text{W} = 1\text{W}$$

   (b) In terms of $|\Gamma|$ at the antenna, how much power is reflected back toward the PA?

   $P_{ref} = |\Gamma|^2\, P_{fwd}$.

   (c) Find the largest $|\Gamma|$ the system can tolerate before the diode is destroyed.

   $$|\Gamma|_{max} = \sqrt{\frac{P_{max}}{P_{fwd}}} = \sqrt{0.1} \approx 0.316$$

   (d) Convert to a VSWR limit and a return-loss limit, and state the datasheet rule.

   $$\text{VSWR}_{max} = \frac{1.316}{0.684} \approx 1.93, \qquad \text{RL}_{min} = -10\log_{10}(0.1) = 10\text{ dB}$$

   Rule: keep VSWR $\leqslant 1.9$ (return loss $\geqslant 10$ dB); reflected power must stay $\geqslant 10$ dB below forward.

   (e) In one sentence: how does a circulator or isolator between the PA and antenna change this picture?

   It routes the reflected wave into a matched dump-port load instead of back into the PA, so the diode no longer sees it — the antenna can then run at much higher VSWR (paid for by insertion loss, size, and bandwidth).

Documentation: ______________________________